

Linear systems and elimination

Sharp growth for unequal lower and upper bandwidths

Difficulty: challenging **Importance:** interesting to specialist**Status:** Solved

Independently reviewed resolution - 2026-09-11

Sharp growth classification. Theorem 1 and Sections 2-4 prove $G(0, q) = 1$ and $G(p, q) = h_{p+q}$ for $p \geq 1$, where $h_t = 0$ for $t \leq 0$ and $h_t = 1 + \sum_{r=1}^p h_{t-r}$ otherwise. The upper bound covers every complex input and admissible tie path, with growth over all active entries in the fixed original ordering. A real nonsingular matrix of order $2p + q + 1$ attains it, including zero upper bandwidth.

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The difficulty, importance and rating rationale below are historical assessments of the original open target. Original statements, references and dated audits are preserved.

Rating rationale: Challenging reflects interacting fill-in and pivot choices across arbitrary unequal bandwidths; specialist impact is a sharp stability classification for banded elimination.

Problem statement

Work in exact arithmetic over $\mathbb{C}$. For integers $p, q \geq 0$, define

$$\mathcal{B}_n(p, q) = \{A \in \mathbb{C}^{n \times n} : \det A \neq 0, \quad a_{ij} = 0 \text{ if } i - j > p \text{ or } j - i > q\}.$$

Gaussian elimination with partial pivoting (GEPP) chooses a largest-modulus entry in the active first column, moves its row to the first position, and forms the trailing Schur complement. For a permitted path π , let $S_1 = A, \dots, S_n$ be its active matrices and define

$$\rho(A, \pi) = \frac{\max_{1 \leq k \leq n} \|S_k\|_{\max}}{\|A\|_{\max}}, \quad \|M\|_{\max} = \max_{i,j} |m_{ij}|.$$

Determine, for all $p \neq q$, the sharp bound independent of dimension,

$$G(p, q) = \sup_{n \geq 1 + \max(p, q)} \sup_{A \in \mathcal{B}_n(p, q)} \sup_{\pi \text{ permitted by GEPP}} \rho(A, \pi).$$

Identify the least universal upper bound and matching examples approaching it. All tie choices are included. The bandwidth restrictions apply in the given ordering; no reordering before GEPP is allowed. Bandwidths are *at most* p and q ; unequal pairs with one zero bandwidth are included. For $p \geq 1$, the equal-bandwidth case is known:

$$G(p, p) = 2^{2p-1} - (p-1)2^{p-2}.$$

This asks how unequal bandwidths affect worst-case element growth in banded linear solves. All unequal pairs constitute one problem.

References

N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, second edition, SIAM (2002), Problem 9.15(a), p. 193; definitions and Theorem 9.11, pp. 172–173. Z. Bohte, *Bounds for Rounding Errors in the Gaussian Elimination for Band Systems*, J. Inst. Maths. Applics. 16 (1975), 133–142.

Earlier status check — 2026-09-08

The book’s problem was checked directly. Its wording omits the field; this entry adopts $\mathbb{C}$ from its referenced Theorem 9.11. Searches for *Higham 9.15 growth, growth factor lower bandwidth upper bandwidth*, and *Bohte Gaussian 1975* found no general unequal-bandwidth solution. Shah–Urschel’s [August 31, 2026 revision](#), Theorems 2.2–2.3, studies sparsity constraints without determining this extremal function. No recent explicit reaffirmation of this particular question’s openness was located; the status rests on the original question and this bounded later-literature check.

Audit update — 2026-09-10

Higham’s [Problem 9.15\(a\)](#), p. 193, remains the explicit historical source. Unequal-bandwidth and sharp-growth searches, including the [August 2026 complete-pivoting manuscript](#), did not identify a sharp answer for this partial-pivoting sparsity pattern. This is a bounded historical-source assessment, without a recent explicit reaffirmation.